TIMILEHIN ADEDAYO

GitHub • HuggingFace • adedayotimi123@gmail.com • (+234) 90-6781-2179 • Portfolio • Ibadan • LinkedIn

Results-driven AI/ML Developer with demonstrated expertise in designing and deploying machine learning solutions, including fine-tuned Large Language Models, computer vision systems, and RAG pipelines. Proven track record leading technical teams and optimizing AI-powered features from conception to production. Skilled in Python, TensorFlow, PyTorch, and cloud technologies with a passion for solving complex problems through innovative AI applications.

EDUCATION

Babcock University — **Ogun, Nigeria**

B.Sc Computer Science — **Expected Grad July 2026**

Relevant Coursework: Database Management Systems, Discrete Mathematics, General Statistics, Statistical Methods I & II

TECHNICAL SKILLS

AI/ML Development: LLM Fine-tuning, RAG Architecture, Computer Vision, NLP, Fraud Detection, Prompt Engineering, Cypher (Neo4j).

Programming: Python (FastAPI, SQLAlchemy, Pydantic), JavaScript (React, Node.js), C, Java, SQL.

Tools: Neo4j Aura, PostgreSQL, Docker, TensorFlow, PyTorch, Scikit-Learn, GitHub, Vercel, Railway.

FEATURED PROJECTS

Babcock University Knowledge Graph Ecosystem — March 2026

- Engineered a production-grade RAG platform integrating LLMs with a Neo4j Knowledge Graph (10,000+ academic entities) for high-precision university data retrieval.
- Developed gdg-bu-kg, a published Python SDK with hybrid JWT authentication, reducing query latency by ~500ms and ensuring type-safe interaction.
- Designed a real-time "Command Center" UI using React/Vite and FastAPI, featuring comprehensive telemetry and audit-driven analytics.
- Implemented a dialect-aware database engine with automatic PostgreSQL/SQLite failover for robust production deployment.

Stack: Neo4j Aura, FastAPI, React, Python SDK (Pydantic v2), SQLAlchemy 2.0, PostgreSQL.

BRAIN (Breast Retrieval-Augmented Intelligence Network) — Oct 2025

- Developed an explainable AI system for cancer diagnosis using Swin Transformer (Swin-B) architecture, achieving 100% accuracy on a 4,916 image test set.
- Integrated a retrieval-augmented verification module using Facebook's FAISS library to cross-check predictions against a 24,000+ image knowledge base.
- Utilized Grad-CAM++ and Attention Rollout to provide visual heatmaps of pathological features for transparent clinical decision support.

Stack: Python, PyTorch, FAISS, FastAPI, React, Grad-CAM++.

Fine-tuned Dolphin LLM (Safety Research) — Oct 2025

- Performed memory-efficient fine-tuning on large models to test the effects of LLMs without guardrails in a university setting.

- Optimized data alignment prioritizing honesty and directness for accurate instruction-following using PEFT and Transformers.

Stack: Python, PyTorch, HF (PEFT, Transformers, Accelerate).

AI Job Skill Analyzer — 2025

- Developed an AI-powered system to analyze job descriptions and map required skills to candidate profiles using NLP techniques.

Stack: Python, NLP, FastAPI.

CaseBud | Team Project — May 2025

- Worked in a team of three to tackle SDG 16 by developing a legal chatbot with 100+ users.

Stack: Python, AI/ML (NLP), Web Technologies.

Owltech | Wema Hackathon — Oct 2025

- Automated security tool to detect high-risk "Shadow IT" apps and malicious URLs.
- Developed a FastAPI microservice aggregating real-time data from 5+ external security APIs.

Stack: Python, FastAPI, Docker, Render, HF Inference API.

EXPERIENCE

Co-Founder & CTO — Feb 2026 – Present

Acturian — **Full-time**

- Architect and maintain the core scoring engine, designing ML pipelines that evaluate risk signals across structured and unstructured data sources.
- Integrate fine-tuned transformer models into the scoring pipeline to improve classification accuracy by ~18% over the prior rule-based system.
- Build FastAPI microservices to expose scoring endpoints consumed by internal underwriting tools with sub-100ms p95 latency.
- Collaborate with actuarial analysts to translate domain expertise into feature engineering strategies and model evaluation criteria.
- Own model monitoring and drift detection, maintaining dashboards tracking data distribution shifts and prediction confidence over time.

Founder & AI Lead — Sep 2025 – Present

Cortex AI — **Full-time**

- Designed and shipped a diagnostic inference engine combining computer vision (Swin Transformer) with a RAG-grounded knowledge base of clinical guidelines.
- Built the end-to-end ML pipeline from data ingestion to model serving, supporting real-time inference at scale.
- Developed a lightweight model quantization pipeline enabling deployment on low-bandwidth, low-compute edge devices.
- Implemented data governance protocols ensuring HIPAA-aligned anonymisation and consent management for all patient data.
- Coordinated a cross-functional team of engineers, clinicians, and UX researchers to ship the product on schedule.

Software Developer Intern — Jan – Jun 2025

International Institute of Tropical Agriculture (IITA) — **Internship**

- Built convolutional neural networks for early detection of cassava mosaic disease and brown streak virus from field imagery, achieving 94% validation accuracy.

- Developed automated crop yield estimation pipelines using satellite imagery and Random Forest regressors across multi-season datasets.
- Integrated drone-captured multispectral data into a soil health classification model to guide precision fertiliser application.
- Created a Flask-based internal dashboard for researchers to upload leaf samples and receive real-time disease predictions.
- Collaborated with agronomists to curate and label a 12,000-image dataset of pest-affected maize crops for downstream model training.

Machine Learning Intern **Sep – Dec 2024**
Synthea Labs **Internship**

- Trained and evaluated generative adversarial networks for synthetic tabular data augmentation, reducing class imbalance in fraud detection datasets by 3x.
- Built an automated feature selection pipeline using mutual information and SHAP values, cutting model training time by 40%.
- Deployed inference endpoints on AWS Lambda behind API Gateway, handling 2,000+ daily prediction requests.
- Wrote comprehensive model cards and experiment reports documenting hyperparameter sweeps across 50+ configurations.

ML Engineering Intern **May – Aug 2024**
Meridian Analytics **Internship**

- Developed time-series forecasting models (LSTM, Prophet) for energy consumption prediction across commercial building portfolios.
- Designed a real-time anomaly detection system using Isolation Forest and autoencoders, flagging equipment failures 6 hours ahead of traditional thresholds.
- Optimised ETL pipelines ingesting 500GB+ of sensor telemetry data into a feature store built on DuckDB and Parquet.
- Presented weekly model performance reviews to the engineering leadership team, translating metrics into actionable product insights.

NLP Research Intern **Jan – Apr 2024**
Kova Research **Internship**

- Fine-tuned multilingual BERT models for low-resource African language sentiment analysis, achieving state-of-the-art F1 scores on Yoruba and Igbo benchmarks.
- Built a named entity recognition pipeline for legal document processing, extracting structured data from 10,000+ unstructured contracts.
- Implemented a retrieval-augmented generation prototype using FAISS and GPT-3.5 for internal knowledge base Q&A.
- Authored a technical report on cross-lingual transfer learning strategies, contributing to the team's publication pipeline.

Computer Vision Intern **Jul – Dec 2023**
DeepFrame Systems **Internship**

- Trained YOLOv8 object detection models for real-time PPE compliance monitoring on construction sites, reaching 91% mAP.
- Built a video analytics pipeline processing 30fps RTSP streams with OpenCV, performing frame sampling, tracking, and alert dispatch.
- Developed a Grad-CAM++ visualisation tool for model interpretability, enabling safety officers to audit detection decisions.

- Reduced model inference latency by 35% through TensorRT optimisation and INT8 quantisation on NVIDIA Jetson edge devices.

LEADERSHIP & VOLUNTEERING

GDG Babcock University | Project Manager — **Sep 2025 – Present**

- Lead the developer team in executing complex AI projects, including the University Knowledge Graph Ecosystem. Responsible for project architecture, roadmap planning, and cross-functional team coordination.

GDG Babcock University | Data & AI Track Lead — **Sep 2025 – Present**

- Organise and host technical training sessions for ML careers, engaging 100+ students and mentoring upcoming AI developers through workshops and hands-on sessions.

Babcock Tech Hub | AI Technical Mentor — **2025**

- Served as the primary AI technical mentor, guiding students through hands-on machine learning projects covering computer vision, NLP, and model deployment.
- Conducted weekly office hours and code reviews for mentees building their first end-to-end ML pipelines.

Babcock University 25th Anniversary Hackathon Winner — **October 2025**

- Awarded first place for "BRAIN", an AI-powered diagnostic network for breast cancer retrieval.

Babcock University Computer Club Dev Team Member — **Nov 2025 – Present**

TRAINING & CERTIFICATION

- NVIDIA: Building RAG Agents with LLMs, Fundamentals of Accelerated Data Science, Fundamentals of Deep Learning
- COURSERA: Neural Networks and Deep Learning
- ZERO TO MASTERY: TensorFlow for Deep Learning
- UDEMY: Data Structures Using Python, Complete AI, ML and Data Science Bootcamp